Equality conditions for the teacher-optimal bound

GPT-6 (Codex)

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Scope and conclusion

This report audits the inequalities used to obtain `prop-teacher-opt-main-bound` in `paper.typ`, including the supporting lemmas and their appendix proofs. Later convergence relaxations are outside this proposition's proof.

When $e^{(k)} > 0$ and $\tau > 0$, no system satisfying the assumptions below makes the entire proof tight. The signed equality conditions for the clean-row and selected-row teacher biases conflict. In addition, the final spectral relaxation would require $A_{\mathcal{C}}$ to have rank one. Exact equality is possible when the teacher is perfect or the current iterate is already correct; examples follow.

Notation and assumptions

Fix an iteration k, and use the paper's notation. The iterate error vector is $\boldsymbol{e}^{(k)} = \boldsymbol{x}^{(k)} - \boldsymbol{x}^*$, with $e^{(k)} = \left\|\boldsymbol{e}^{(k)}\right\|_2^2$. The teacher error vector is $\hat{\boldsymbol{x}} - \boldsymbol{x}^*$, and $\tau = \left\|\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon}^*\right\|_\infty$ as in `eq-teacher-err`. For every row i,

$$\begin{aligned} \varepsilon_i^{(k)} - \varepsilon_i^* &= \boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{e}^{(k)}, \quad \hat{\varepsilon}_i - \varepsilon_i^* = \boldsymbol{a}_i^{\mathrm{T}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*), \\ M_i^{(k)} &= \left(\varepsilon_i^{(k)} - \varepsilon_i^*\right)^2 - (\varepsilon_i^*)^2, \\ \delta_i &= 2\varepsilon_i^{(k)}(\varepsilon_i^* - \hat{\varepsilon}_i), \quad \widehat{M}_i^{(k)} = M_i^{(k)} + \delta_i. \end{aligned} \tag{1}$$

The selected index is $\hat{\imath}$, as in `eq-ihat`. The exact Kaczmarz update gives

$$e^{(k+1)} = e^{(k)} - M_{\hat{\imath}}^{(k)}. \tag{2}$$

We assume real matrices, unit rows, $n > d$, $|\mathcal{C}| > d$, $\varepsilon_j^* = 0$ for $j \in \mathcal{C}$, and full column rank of $A_{\mathcal{C}}$. The paper states that $|\mathcal{C}| > d$ implies full column rank, but that implication is false; full rank is a separate required assumption. It makes $\tau = 0$ equivalent to $\hat{\boldsymbol{x}} = \boldsymbol{x}^*$.

All equality conditions below concern a single iteration. Equality for one step need not persist along a trajectory.

Audit of the selected-row residual bound

The appendix proof of `eq-delta-bound` bounds $|\delta_i|$. Each step can be tight individually, with the following conditions.

1. **Triangle inequality:** $\left|\varepsilon_i^{(k)}\right| \le \left|\varepsilon_i^{(k)} - \varepsilon_i^*\right| + |\varepsilon_i^*|$. Equality holds exactly when $\left(\varepsilon_i^{(k)} - \varepsilon_i^*\right)\varepsilon_i^* \ge 0$.
2. **Row Cauchy–Schwarz:** $\left|\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{e}^{(k)}\right| \le \left\|\boldsymbol{e}^{(k)}\right\|_2 = \sqrt{e^{(k)}}$. If $e^{(k)} > 0$, equality holds exactly when $\boldsymbol{a}_i = \pm\frac{\boldsymbol{e}^{(k)}}{\sqrt{e^{(k)}}}$; if $e^{(k)} = 0$, it is automatic.
3. **Noise maximum:** $|\varepsilon_i^*| \le \left\|\boldsymbol{\varepsilon}^*\right\|_\infty$. Equality means $|\varepsilon_i^*| = \left\|\boldsymbol{\varepsilon}^*\right\|_\infty$.
4. **Absolute product:** $|\delta_i| = 2\left|\varepsilon_i^{(k)}\right||\hat{\varepsilon}_i - \varepsilon_i^*|$ is an identity.
5. **Teacher maximum:** $|\hat{\varepsilon}_i - \varepsilon_i^*| \le \tau$. In the product above, this step is tight if this absolute value is τ or $\varepsilon_i^{(k)} = 0$.

6. **Combined estimate:** if $\tau > 0$ and $\sqrt{e^{(k)}} + \|\boldsymbol{\varepsilon}^*\|_\infty > 0$, equality in $|\delta_i| \le 2\big(\sqrt{e^{(k)}} + \|\boldsymbol{\varepsilon}^*\|_\infty\big)\tau$ holds exactly when

$$\begin{aligned} \big|\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{e}^{(k)}\big| &= \sqrt{e^{(k)}}, \quad |\varepsilon_i^*| = \|\boldsymbol{\varepsilon}^*\|_\infty, \\ \big(\boldsymbol{a}_i^{\mathrm{T}}\boldsymbol{e}^{(k)}\big)\varepsilon_i^* &\ge 0, \quad |\hat{\varepsilon}_i - \varepsilon_i^*| = \tau. \end{aligned} \tag{3}$$

   If either multiplying factor on the right is zero, both sides vanish. A one-dimensional unit row can attain this residual estimate, but need not be selected by the teacher criterion.

Selection, averaging, and signs

1. **Teacher maximization:** for any comparison row j,

$$M_{\hat{\imath}}^{(k)} = \widehat{M}_{\hat{\imath}}^{(k)} - \delta_{\hat{\imath}} \ge \widehat{M}_j^{(k)} - \delta_{\hat{\imath}} = M_j^{(k)} + \delta_j - \delta_{\hat{\imath}}. \tag{4}$$

   Equality holds exactly when $\widehat{M}_j^{(k)} = \widehat{M}_{\hat{\imath}}^{(k)}$.

2. **Averaging over clean rows:** since $\varepsilon_j^* = 0$ on $\mathcal{C}$,

$$M_{\hat{\imath}}^{(k)} \ge \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\big(\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{e}^{(k)}\big)^2 + \frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\delta_j - \delta_{\hat{\imath}}. \tag{5}$$

   Equality holds exactly when every clean row ties the selected row in teacher score: $\forall j \in \mathcal{C}, \widehat{M}_j^{(k)} = \widehat{M}_{\hat{\imath}}^{(k)}$. The gap is the average of the nonnegative differences between those teacher scores.

3. **Clean-bias absolute value:** replacing the clean average bias by its negative absolute value is tight exactly when $\sum_{j\in\mathcal{C}}\delta_j \le 0$.

4. **Selected-bias absolute value:** replacing $-\delta_{\hat{\imath}}$ by $-|\delta_{\hat{\imath}}|$ is tight exactly when $\delta_{\hat{\imath}} \ge 0$, or equivalently $\Big(\varepsilon_{\hat{\imath}}^{(k)}\Big)(\hat{\varepsilon}_{\hat{\imath}} - \varepsilon_{\hat{\imath}}^*) \le 0$.

These sign conditions are essential to simultaneous equality.

Gain and clean teacher bias

1. **Smallest singular value:** the gain bound is

$$\frac{1}{|\mathcal{C}|}\sum_{j\in\mathcal{C}}\big(\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{e}^{(k)}\big)^2 = \frac{\big\|A_{\mathcal{C}}\boldsymbol{e}^{(k)}\big\|_2^2}{|\mathcal{C}|} \ge \alpha e^{(k)}, \tag{6}$$

   where $\alpha = \frac{\sigma^2_{\min(A_{\mathcal{C}})}}{|\mathcal{C}|}$ is defined in `eq-master-ineq`. If $e^{(k)} > 0$, equality holds exactly when $\boldsymbol{e}^{(k)}$ is a right singular vector for the smallest singular value. If $e^{(k)} = 0$, equality is automatic.

2. **Clean Cauchy–Schwarz:**

$$\Big|\big(A_{\mathcal{C}}\boldsymbol{e}^{(k)}\big)^{\mathrm{T}}A_{\mathcal{C}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\Big| \le \big\|A_{\mathcal{C}}\boldsymbol{e}^{(k)}\big\|_2\|A_{\mathcal{C}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\|_2. \tag{7}$$

   Equality holds if either vector is zero or they are real scalar multiples. Under full column rank, if $e^{(k)} > 0$ and $\tau > 0$, the teacher error must be a nonzero scalar multiple of $\boldsymbol{e}^{(k)}$. Equality with the required nonnegative sign of the clean bias requires a positive multiple.

3. **Euclidean and maximum norms:** $\|A_{\mathcal{C}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\|_2 \le \sqrt{|\mathcal{C}|}\|A_{\mathcal{C}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\|_\infty$. Equality holds exactly when all clean-row teacher residual errors have equal absolute value. In a product with $\big\|A_{\mathcal{C}}\boldsymbol{e}^{(k)}\big\|_2 = 0$, the product vanishes without requiring equality here.

4. **Largest singular value and clean restriction:** $\left\|A_{\mathcal{C}} e^{(k)}\right\|_2 \le \sigma_{\max(A_{\mathcal{C}})}\sqrt{e^{(k)}}$ and $\left\|A_{\mathcal{C}}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\right\|_\infty \le \tau$. For nonzero factors, equality requires respectively a largest right singular vector and a clean row attaining the global teacher-error maximum. Equality in the preceding maximum-norm comparison additionally requires every clean row to attain that maximum.

5. **Simultaneous smallest and largest singular values:** for $e^{(k)} > 0$, simultaneous equality in the gain bound and the largest-singular-value bound requires

$$\sigma_{\min(A_{\mathcal{C}})} = \sigma_{\max(A_{\mathcal{C}})}, \quad A_{\mathcal{C}}^{\mathrm{T}} A_{\mathcal{C}} = \left(\frac{|\mathcal{C}|}{d}\right) I_d. \tag{8}$$

   The second identity follows from full column rank and row normalization, which gives $\operatorname{tr}\left(A_{\mathcal{C}}^{\mathrm{T}} A_{\mathcal{C}}\right) = |\mathcal{C}|$.

6. **Final spectral relaxation:** $\sigma_{\max(A_{\mathcal{C}})} \le \|A_{\mathcal{C}}\|_F = \sqrt{|\mathcal{C}|}$. Equality in the inequality holds exactly when $A_{\mathcal{C}}$ has rank one. Under full column rank this means $d = 1$. In the bias product the relaxation is also tight when $\tau\sqrt{e^{(k)}} = 0$, since then the product vanishes regardless of the matrix inequality.

The primitive clean norm bounds can all be attained at once: take $d = 1$, every clean row equal to 1, and positive scalar iterate and teacher errors. That example does not satisfy the selected-bias sign condition at a row saturating `eq-delta-bound`. For $d > 1$, repeated coordinate rows attain the smallest and largest singular-value bounds together, but the Frobenius relaxation is strict when $\tau > 0$ and $e^{(k)} > 0$.

Simultaneous equality when both errors are positive

Suppose $e^{(k)} > 0$ and $\tau > 0$. Equality in the proposition's derivation requires clean rows to tie the selected teacher score, the clean average bias to be nonpositive, and $\delta_{\hat{\imath}} \ge 0$. It also requires simultaneous equality in the gain and largest-singular-value bounds, equality in clean Cauchy–Schwarz, all clean teacher residual errors to attain absolute value τ, and equality in `eq-delta-bound` at $\hat{\imath}$.

Already the signed conditions make this impossible. Equality in clean Cauchy–Schwarz with the clean bias sign requires $\hat{\boldsymbol{x}} - \boldsymbol{x}^* = \lambda \boldsymbol{e}^{(k)}$ for $\lambda > 0$. Equality in `eq-delta-bound` at $\hat{\imath}$ gives $\left|\boldsymbol{a}_{\hat{\imath}}^{\mathrm{T}} \boldsymbol{e}^{(k)}\right| = \sqrt{e^{(k)}}$ and $\left(\boldsymbol{a}_{\hat{\imath}}^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)\varepsilon_{\hat{\imath}}^* \ge 0$. Therefore

$$\left(\varepsilon_{\hat{\imath}}^{(k)}\right)(\hat{\varepsilon}_{\hat{\imath}} - \varepsilon_{\hat{\imath}}^*) = \left(\boldsymbol{a}_{\hat{\imath}}^{\mathrm{T}} \boldsymbol{e}^{(k)} + \varepsilon_{\hat{\imath}}^*\right)\left(\lambda \boldsymbol{a}_{\hat{\imath}}^{\mathrm{T}} \boldsymbol{e}^{(k)}\right) > 0, \tag{9}$$

so $\delta_{\hat{\imath}} < 0$. This contradicts the selected-bias equality requirement $\delta_{\hat{\imath}} \ge 0$. Since each relaxation contributes a nonnegative gap to the lower bound for $M_{\hat{\imath}}^{(k)}$, the contradiction proves that the final inequality `eq-master-ineq` is strict whenever $e^{(k)} > 0$ and $\tau > 0$.

6.1 A near-sharp family with positive iterate and teacher errors

Although equality is impossible in this regime, the bound can be arbitrarily close in relative terms. Let $R > 0$ and $s > 0$, and take $d = 2$, $n = 5$, with rows and data

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{x}^* = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2s \end{pmatrix}, \tag{10}$$

where the first four rows are clean and the fifth is corrupted. In the paper's residual convention $\boldsymbol{\varepsilon}^* = A\boldsymbol{x}^* - \boldsymbol{b}$, so

$$\boldsymbol{\varepsilon}^* = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -2s \end{pmatrix}, \quad \hat{\boldsymbol{x}} = \begin{pmatrix} s \\ s \end{pmatrix}, \quad \boldsymbol{x}^{(k)} = \begin{pmatrix} R \\ R \end{pmatrix}. \tag{11}$$

Thus $\tau = \|A(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)\|_\infty = s > 0$, $B \coloneqq \|\boldsymbol{\varepsilon}^*\|_\infty = 2s$, and $e^{(k)} = 2R^2 > 0$. The clean frame has $A_{\mathcal{C}}^{\mathrm{T}} A_{\mathcal{C}} = 2I_2$, hence $\alpha = \frac{2}{4} = \frac{1}{2}$.

The current residual and teacher residual vectors are

$$\boldsymbol{\varepsilon}^{(k)} = A\boldsymbol{x}^{(k)} - \boldsymbol{b} = \begin{pmatrix} R \\ R \\ R \\ R \\ R - 2s \end{pmatrix}, \setminus \hat{\boldsymbol{\varepsilon}} = A\hat{\boldsymbol{x}} - \boldsymbol{b} = \begin{pmatrix} s \\ s \\ s \\ s \\ -s \end{pmatrix}. \tag{12}$$

Every clean row has teacher score $\widehat{M}_j^{(k)} = R^2 - 2Rs$. The fifth row ties it because

$$\widehat{M}_5^{(k)} = (R - 2s)^2 - 2(R - 2s)(-s) = R^2 - 2Rs. \tag{13}$$

Therefore choosing the fifth row is permitted by the argmax rule (the example uses a tie resolved in favor of that row). Its Kaczmarz update is $\boldsymbol{x}^{(k+1)} = \begin{pmatrix} 2s \\ R \end{pmatrix}$, and hence

$$e^{(k+1)} = R^2 + 4s^2. \tag{14}$$

The right-hand side of `eq-master-ineq` is

$$(1 - \alpha)e^{(k)} + 4\tau\sqrt{e^{(k)}} + 2B\tau = R^2 + 4\sqrt{2}Rs + 4s^2. \tag{15}$$

So the inequality is strict, with additive gap $4\sqrt{2}Rs$, but

$$\texttt{frac}\left\{e^{(k+1)}\right\}\left\{(1 - \alpha)e^{(k)} + 4\tau\sqrt{e^{(k)}} + 2B\tau\right\} = \texttt{frac}\left\{R^2 + 4s^2\right\}\left\{R^2 + 4\sqrt{2}Rs + 4s^2\right\} \to 1 \quad \text{as} \quad \frac{s}{R} \to 0. \tag{16}$$

This is a meaningful worst-case sense in which the bound is close: for every member of the family both errors are nonzero, yet the relative slack can be made arbitrarily small. The closeness is multiplicative; the additive slack itself is not zero for finite $R, s > 0$. The construction also makes the dependence on tie-breaking explicit: if a clean row is selected instead, its update gives $e^{(k+1)} = R^2$, below the displayed near-worst-case value.

Degenerate cases and exact equality examples

7.1 Perfect teacher: $\tau = 0$

Full column rank implies $\hat{\boldsymbol{x}} = \boldsymbol{x}^*$. Equality in `eq-master-ineq` holds exactly when

$$\begin{aligned} A_{\mathcal{C}}^{\mathrm{T}} A_{\mathcal{C}} \boldsymbol{e}^{(k)} &= \sigma_{\min(A_{\mathcal{C}})}^2 \boldsymbol{e}^{(k)}, \\ \forall j \in \mathcal{C}, \left(\boldsymbol{a}_j^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)^2 &= \alpha e^{(k)}, \\ \max_{1 \le i \le n} \left(\left(\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)^2 - \left(\varepsilon_i^*\right)^2\right) &= \alpha e^{(k)}. \end{aligned} \tag{17}$$

When $e^{(k)} = 0$, the first condition is automatic and the maximum is zero because there is a clean row. For positive error, these conditions say the iterate error is in the smallest singular subspace, every clean row attains the same gain, and no row has a larger criterion.

An exact example in any dimension uses two copies of each coordinate row, $n = |\mathcal{C}| = 2d$, all rows clean, $\boldsymbol{x}^* = \hat{\boldsymbol{x}} = \boldsymbol{b} = \boldsymbol{\varepsilon}^* = \mathbf{0}$, and $\boldsymbol{x}^{(k)} = q(1, ..., 1)^{\mathrm{T}}$ for $q > 0$. Then $A_{\mathcal{C}}^{\mathrm{T}} A_{\mathcal{C}} = 2I_d$, $\alpha = \frac{1}{d}$, and every row criterion is q^2. Every selected update leaves squared error $(d-1)q^2 = (1-\alpha)e^{(k)}$, attaining the proposition.

7.2 Correct iterate and imperfect teacher: $e^{(k)} = 0$, $\tau > 0$

The proposition becomes $e^{(k+1)} \le 2\|\boldsymbol{\varepsilon}^*\|_\infty \tau$. If $\|\boldsymbol{\varepsilon}^*\|_\infty = 0$, all current residuals vanish, all criteria tie at zero, and equality holds for any selection.

If $\|\boldsymbol{\varepsilon}^*\|_\infty > 0$, equality holds exactly when the selected row satisfies

$$\begin{aligned} |\varepsilon_{\hat{\imath}}^*| &= \|\boldsymbol{\varepsilon}^*\|_\infty = 2\tau, \\ |\hat{\varepsilon}_{\hat{\imath}} - \varepsilon_{\hat{\imath}}^*| &= \tau, \\ \varepsilon_{\hat{\imath}}^*(\hat{\varepsilon}_{\hat{\imath}} - \varepsilon_{\hat{\imath}}^*) &= -\|\boldsymbol{\varepsilon}^*\|_\infty \tau, \end{aligned} \tag{18}$$

and every row has $\widehat{M}_i^{(k)} \le 0$. Then the clean rows and the selected row tie at zero. Necessity follows from equality in the selected-row residual bound and the selected-bias sign step; conversely these conditions give equality directly.

For an example, let $d = 1$, $n = 3$, $\mathcal{C} = \{1, 2\}$, $A = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, $\boldsymbol{x}^* = \boldsymbol{x}^{(k)} = 0$, $\hat{\boldsymbol{x}} = T$ for $T > 0$, $\boldsymbol{\varepsilon}^* = \begin{pmatrix} 0 \\ 0 \\ -2T \end{pmatrix}$, and $\boldsymbol{b} = \begin{pmatrix} 0 \\ 0 \\ 2T \end{pmatrix}$. Then $\tau = T$, $\|\boldsymbol{\varepsilon}^*\|_\infty = 2T$, and all teacher criteria are zero. Selecting the third row is allowed; the update gives $e^{(k+1)} = 4T^2 = 2\|\boldsymbol{\varepsilon}^*\|_\infty \tau$. Thus exact equality can depend on how a tie is resolved.

These cases cover all possibilities for whether $e^{(k)}$ and τ vanish.

Bounds that keep the paper's notation

8.1 Keep the clean largest singular value

Omitting the final spectral-to-Frobenius relaxation gives

$$\begin{aligned} M_{\hat{\imath}}^{(k)} &\ge \alpha e^{(k)} - 2\tau\left(1 + \frac{\sigma_{\max(A_{\mathcal{C}})}}{\sqrt{|\mathcal{C}|}}\right)\sqrt{e^{(k)}} - 2\|\boldsymbol{\varepsilon}^*\|_\infty \tau, \\ e^{(k+1)} &\le (1-\alpha)e^{(k)} + 2\tau\left(1 + \frac{\sigma_{\max(A_{\mathcal{C}})}}{\sqrt{|\mathcal{C}|}}\right)\sqrt{e^{(k)}} + 2\|\boldsymbol{\varepsilon}^*\|_\infty \tau. \end{aligned} \tag{19}$$

For full-rank $A_{\mathcal{C}}$ with $d > 1$, $\sigma_{\max(A_{\mathcal{C}})} < \sqrt{|\mathcal{C}|}$, so this improves the displayed bound in `eq-master-ineq` when $\tau\sqrt{e^{(k)}} > 0$. Also,

$$\frac{\sigma_{\max(A_{\mathcal{C}})}^2}{|\mathcal{C}|} \le 1 - (d-1)\alpha, \tag{20}$$

because the other $d-1$ squared singular values are at least $\sigma_{\min(A_{\mathcal{C}})}^2$.

8.2 Preserve the clean-row signs

The selection rule itself yields the exact lower bound

$$\begin{aligned} M_{\hat{\imath}}^{(k)} \ge \max_{j \in \mathcal{C}} \Big(&\left(\boldsymbol{a}_j^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)^2 - 2\left(\boldsymbol{a}_j^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)\left(\boldsymbol{a}_j^{\mathrm{T}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)}\right)\Big) \\ &+ 2\left(\varepsilon_{\hat{\imath}}^{(k)}\right)(\hat{\varepsilon}_{\hat{\imath}} - \varepsilon_{\hat{\imath}}^*). \end{aligned} \tag{21}$$

The first comparison is tight when the selected row ties a maximizing clean row; averaging the clean comparisons recovers the proof's clean-gain term. This form keeps the selected sign information that the absolute-value estimates discard.

The exact update also gives the teacher-independent bound

$$e^{(k+1)} = e^{(k)} - \left(\boldsymbol{a}_{\hat{i}}^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)^2 + (\varepsilon_{\hat{i}}^*)^2 \le e^{(k)} + \|\boldsymbol{\varepsilon}^*\|_\infty^2. \tag{22}$$

One can take the minimum of this and `eq-master-ineq`.

8.3 A sharp one-dimensional bound

In $d = 1$, every normalized row is ± 1. Thus $\left(\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)^2 = e^{(k)}$ and $\left(\boldsymbol{a}_i^{\mathrm{T}} \boldsymbol{e}^{(k)}\right)\left(\boldsymbol{a}_i^{\mathrm{T}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)}\right) = \boldsymbol{e}^{(k)}(\hat{\boldsymbol{x}} - \boldsymbol{x}^*)$ for every row. Comparing the selected score to a clean row gives

$$(\varepsilon_{\hat{i}}^*)^2 + 2\varepsilon_{\hat{i}}^*(\hat{\varepsilon}_{\hat{i}} - \varepsilon_{\hat{i}}^*) \le 0, \quad |\varepsilon_{\hat{i}}^*| \le 2\tau. \tag{23}$$

The one-dimensional update leaves error $(\varepsilon_{\hat{i}}^*)^2$, so

$$e^{(k+1)} \le \left(\|\boldsymbol{\varepsilon}^*\|_\infty\right)^2, \quad e^{(k+1)} \le 4\tau^2. \tag{24}$$

Hence $e^{(k+1)} \le \min\left(\left(\|\boldsymbol{\varepsilon}^*\|_\infty\right)^2, 4\tau^2\right)$, independent of the current iterate error. This is sharp: the preceding three-row example with $\boldsymbol{x}^{(k)} = R > 0$ instead of zero has all teacher scores tied and attains $e^{(k+1)} = 4T^2$. More generally, choosing the selected row's noise magnitude below or above 2τ attains respectively the first or second term of the minimum, with an additional row if needed to set $\|\boldsymbol{\varepsilon}^*\|_\infty$.

Corrections to the paper's derivation

The clean-bias paragraph begins with an expression involving $\boldsymbol{a}_j^{\mathrm{T}(\varepsilon - \varepsilon^*)}$. These are incompatible dimensions: residual vectors have length n, while $\boldsymbol{a}_j$ has length d. The intended vectors are $\boldsymbol{x} - \boldsymbol{x}^*$ and $\hat{\boldsymbol{x}} - \boldsymbol{x}^*$.

The replacement of ε_j by $\varepsilon_j - \varepsilon_j^*$ is valid because j is clean. State that restriction explicitly; for a general row the noise term remains. Finally, specify whether the selection rule permits arbitrary maximizers or uses a fixed tie rule. Equality for an imperfect teacher requires a corrupted row tied with every clean row, so a sharpness claim should state its dependence on tie resolution.